VECTOR SPACES · RECOGNITION

Subspace or not, and which clause decides

The test has three clauses — contains $\mathbf{0}$, closed under addition, closed under scaling — and a candidate fails by failing one of them. Sorting by which clause is what turns a list of examples into a diagnostic.

8
CANDIDATES

PASSES — BY DEFINITION OR TRIVIALLY 3

1 **The whole space**

the largest subspace

$$V \subseteq V$$

Closure holds because V is a vector space to begin with. Worth stating so that "subspace" is understood as **at most** V rather than strictly smaller — proper subspaces are the special case, not the definition.

2 **The zero subspace**

$\dim = 0$; the smallest

$$\{\mathbf{0}\}$$

All three clauses hold directly: $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $c\mathbf{0} = \mathbf{0}$. Its dimension is zero, not one — the empty set is its basis, since $\mathbf{0}$ is dependent on its own and cannot belong to one.

3 **Any span**

automatically a subspace

$$\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$$

A combination of combinations is a combination, so closure is immediate. This is the cheapest way to prove something is a subspace — exhibit it as a span and the test is already passed. Lines and planes through the origin are the $k = 1$ and $k = 2$ cases.

PASSES — BECAUSE LINEARITY FORCES IT 2

4 **Null space**

subspace of $\mathbb{R}^n$

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

If $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$ then $A(c\mathbf{u} + d\mathbf{v}) = \mathbf{0}$ — one line, both closure clauses. The homogeneous right-hand side is what makes it work, which is exactly why $A\mathbf{x} = \mathbf{b}$ below fails.

5 **Column and row spaces**

spans of the columns and rows

$$\text{Col}(A), \text{Row}(A)$$

Both are spans, so they inherit the result above without further argument. Together with the null space and left null space these are the four fundamental subspaces.

FAILS — AND WHICH CLAUSE 3

6 **Line or plane off the origin**

fails clause 1 — no $\mathbf{0}$

$$\{\mathbf{x}_0 + t\mathbf{v}\},\ \mathbf{x}_0 \notin \text{Span}\{\mathbf{v}\}$$

Affine rather than linear. The origin test is the cheapest of the three and should be run first — it disqualifies this immediately, with no closure argument needed.

7 **Solution set of $A\mathbf{x} = \mathbf{b}$**

fails clause 1 — $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$

$$\mathbf{x}_p + \text{Null}(A),\ \mathbf{b} \neq \mathbf{0}$$

The null space translated off the origin. Not a subspace, but a coset of one — which is why the general solution has the shape it does and why its **dimension** still comes from the null space.

8 **First quadrant of $\mathbb{R}^2$**

fails clause 3 only

$$\{(x, y) : x \geq 0,\ y \geq 0\}$$

Contains the origin and is closed under addition — two clauses out of three. Multiplying by -1 leaves the set, and that alone disqualifies it. The instructive case: passing most of the test is not passing it.

Note the near-misses are near in different ways. An affine set is closed under nothing that matters because it misses the origin; the first quadrant contains the origin and survives addition but not negative scaling. Both are the same distance from being a subspace and fail for unrelated reasons — which is why all three clauses have to be checked rather than the one that looks likeliest.